

COSC349 Assignment 2 (2026): Redesigning and Deploying Software for the Cloud

Project Report: SmartPantry Cloud Infrastructure & Architecture

Project & Submission Metadata

Attribute	Details
Course	

COSC349: Cloud Computing Architecture (2026) |
Assignment	Assignment 2: Redesigning and Deploying Software for the Cloud
Project Title	🍽️ **SmartPantry**: Cloud-Native Pantry Inventory Management System
Author / Student Name	Gunit Chawla
Student Username / ID	`chagu186`
Repository Access	Teaching staff granted access (`dme26` on GitHub / GitLab / Bitbucket)
Shared Repository URL	`https://github.com/gunitchawla/SmartPantry.git`
Submitted Git Commit	`9984909c56e5886690d9b4a326d3249f99224edc`
Screen Recording URL	*(Insert University of Otago OneDrive / YouTube URL)*
Target Cloud Provider	Amazon Web Services (AWS) via AWS Academy Learner Lab (`us-east-1`)

Executive Summary

In Assignment 1, **SmartPantry** was deployed locally as a three-tier system across three virtual machines provisioned with Vagrant and Docker (Frontend, Backend, and a self-hosted MySQL VM).

For Assignment 2, SmartPantry has been fundamentally redesigned into an enterprise-grade, public cloud architecture hosted on Amazon Web Services (AWS). Rather than merely transferring local virtual machines onto Amazon EC2 instances, the cloud deployment purposefully incorporates managed cloud services:

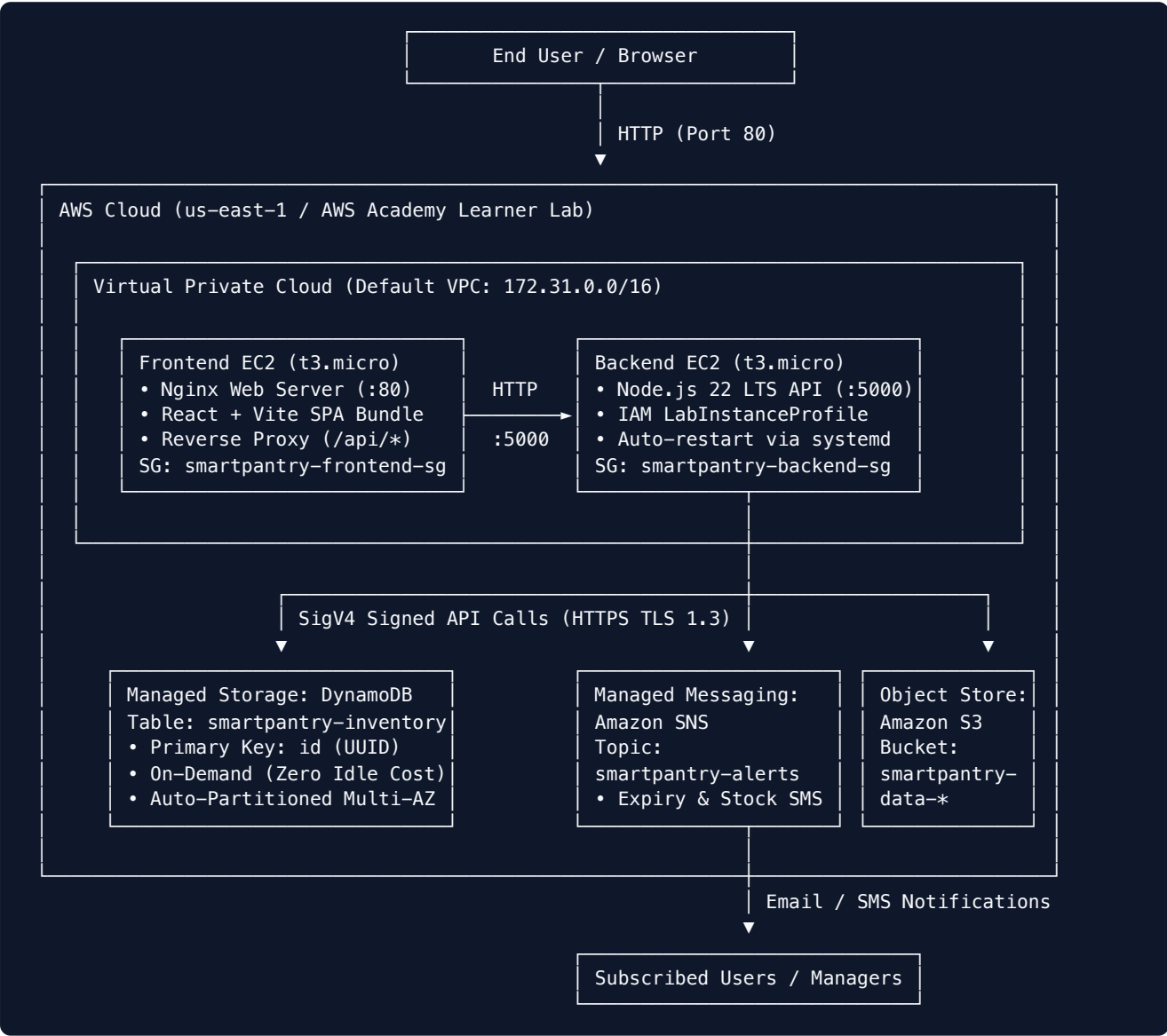
- Compute Tier (Compute Approach 1):** Two decoupled, interacting virtual machines deployed on Amazon EC2—a **Frontend Instance** serving a React Single-Page Application (SPA) behind an **Nginx** reverse proxy on HTTP port 80, and a **Backend Instance** running a Node.js Express REST API on port 5000.
- Primary Managed Storage Service: Amazon DynamoDB**, a serverless NoSQL database providing high-throughput, sub-10ms latency, automatic multi-AZ redundancy, and zero idle cost.
- Further Non-EC2 Managed Service: Amazon Simple Notification Service (SNS)**, providing decoupled, event-driven alerting when items near expiration, expire, or drop below stock thresholds, alongside aggregated inventory audit reports.
- Managed Object Storage & Telemetry: Amazon S3** for immutable inventory snapshot exports and **Amazon CloudWatch** for system telemetry and metric alarms.

5. **Infrastructure as Code (IaC)**: 100% automated, reproducible provisioning via **Terraform** (`>= 1.5.0`), tailored specifically for AWS Academy Learner Lab constraints.

1. Cloud Architecture and Implementation (40%)

1.1. System Overview & Architecture Diagram

SmartPantry decouples the presentation, business logic, storage, and notification layers across a private cloud topology.



1.2. Identification and Justification of Compute Approach

Selected Approach: Approach 1: At least two interacting virtual machines, using EC2 or container instances, together with a further non-EC2 managed cloud service distinct from the storage service.

Technical Justification:

- **Presentation vs Application Decoupling:** Hosting the frontend and backend on distinct EC2 virtual machines strictly enforces separation of concerns. The frontend instance acts as a dedicated presentation gateway, serving pre-compiled static web assets and terminating user HTTP sessions, while the backend instance focuses purely on business logic validation and database operations.

- **Reverse Proxy Architecture:** The frontend Nginx server acts as an internal reverse proxy for all API traffic matching `/api/*`. This architectural pattern eliminates Cross-Origin Resource Sharing (CORS) preflight bottlenecks in production, insulates the backend API from direct exposure to internet crawlers, and allows seamless backend replacement or scaling without altering client-side code.
- **Fail-Safe Process Supervision:** Both instances utilize standard Linux `systemd` unit services (`smartpantry-backend.service` and `nginx.service`), ensuring that any runtime crash results in immediate auto-restart without manual intervention.

1.3. Identification and Justification of Managed Storage Service & Data Model

Selected Storage Service: Amazon DynamoDB (Table: `smartpantry-inventory`).

Technical Justification:

In Assignment 1, SmartPantry used a self-hosted MySQL instance running inside a local VM. When migrating to the cloud, we evaluated **Amazon RDS (MySQL)** versus **Amazon DynamoDB**:

1. **Serverless Provisioning Speed:** In AWS Academy Learner Lab, spinning up an Amazon RDS instance takes 12–15 minutes, requires multi-subnet DB subnet groups, and frequently exhausts student account quotas. DynamoDB tables are provisioned instantaneously (< 5 seconds) via Terraform.
2. **Cost Efficiency:** A continuous `db.t3.micro` RDS instance incurs ongoing hourly fees (\\$0.017/hour plus storage fees), depleting student budgets even when the application is idle. DynamoDB in `PAY_PER_REQUEST` billing mode incurs **\\$0.00 idle cost**.
3. **High Availability by Default:** DynamoDB automatically replicates data synchronously across three Availability Zones (AZs) within the `us-east-1` region without needing complex read-replica or multi-AZ cluster configuration.

Data Model and Access Patterns:

- **Partition Key:** `id` (`String` UUID). This distributes read and write operations uniformly across DynamoDB storage partitions.
- **Attributes:**
 - `name` (`String`): Descriptive name of the grocery/pantry item.
 - `quantity` (`Number`): Units currently in stock.
 - `expiry_date` (`String` ISO 8601 `YYYY-MM-DD`): Expiry date.
 - `category` (`String`): E.g., `Dairy`, `Bakery`, `Pantry`, `Produce`.
 - `status` (`String`): Derived freshness status (`fresh`, `soon`, or `expired`).
 - `created_at` (`String` ISO 8601 timestamp): Audit creation timestamp.
- **Primary Access Patterns:**
 - `Scan` : Retrieve active inventory for the dashboard.
 - `PutItem` : Add new pantry products.
 - `DeleteItem` : Consume or discard pantry products.
 - `UpdateItem` : Adjust remaining quantities upon item usage.

1.4. Identification and Justification of Additional Managed Cloud Services

SmartPantry integrates three further managed AWS cloud services:

1. Amazon Simple Notification Service (SNS) — Primary Further Managed Service

- **Topic:** `smartpantry-alerts`.

- **Justification:** A pantry inventory system must notify users before perishable goods spoil. Rather than having clients continuously poll the server or coupling notification delivery to EC2 compute, SmartPantry delegates event dispatching to Amazon SNS.
- **Coherent Workflow:**
 - When an item is added whose expiry is within 3 days (`status == "soon"`) or already expired, the backend dispatches a high-priority SNS alert.
 - When an item's stock drops to 2 units or fewer, a low-stock alert is published.
 - When an operator clicks **Run Expiry & Stock Audit** in the web dashboard or executes an automated audit, the backend scans all inventory and dispatches an aggregated inventory digest.
 - Users can subscribe their email addresses directly via the UI or API (`POST /products/subscribe`).

2. Amazon Simple Storage Service (S3) — Managed Object Storage

- **Bucket:** `smartpantry-data-${random_id}`.
- **Justification:** Storing point-in-time database dumps or business reports inside DynamoDB is inefficient. Amazon S3 provides cheap, durable (\$99.999999999\%\$), encrypted object storage. SmartPantry implements an automated report generator that serializes current inventory into JSON/CSV snapshots and uploads them to S3 (`POST /products/export`).

3. Amazon CloudWatch — Operational Telemetry & Alarms

- **Alarm:** `smartpantry-backend-high-cpu`.
- **Justification:** CloudWatch continuously monitors the EC2 backend instance's CPU utilization. If average CPU utilization exceeds 80\%\$ across two consecutive 2-minute periods, CloudWatch immediately changes state to `ALARM` and triggers the `smartpantry-alerts` SNS topic.

1.5. Non-Trivial Cloud Architecture Decisions

Decision 1: Database Migration — Amazon DynamoDB (NoSQL) vs. Amazon RDS (MySQL)

- **Context:** Assignment 1 utilized a relational MySQL database. We had to decide whether to adopt Amazon RDS (relational) or Amazon DynamoDB (serverless NoSQL).
- **Alternative Considered: Amazon RDS for MySQL** (`db.t3.micro`).
- **Evaluation & Rationale:** While RDS retains SQL schema compatibility with Assignment 1, it suffers from several severe drawbacks in an academic cloud setting:

1. Long provisioning latency (~12–15 minutes), making rapid Terraform testing tedious.
2. Non-zero idle costs (\\$12–\\$15/month for compute and storage), draining student lab credits.
3. Single-point-of-failure unless expensive Multi-AZ clustering is provisioned.

In contrast, DynamoDB is native to the cloud, provisions in seconds, offers automatic multi-AZ redundancy, requires zero schema migrations when new product attributes are added, and costs **\\$0.00** when idle. We redesigned the data access layer to use DynamoDB DocumentClient.

Decision 2: Decoupled Alerting — Amazon SNS vs. Client-Side Polling / In-App Sockets

- **Context:** The application needed an automated mechanism to alert householders when perishable goods near their expiration date.
- **Alternative Considered:** In-process WebSockets / Server-Sent Events (SSE) or client-side polling.
- **Evaluation & Rationale:** In-process socket connections require stateful connections between the client and EC2 instance, impeding horizontal scalability and draining VM memory. If the user closes the browser tab, notifications are lost. By routing expiry warnings to **Amazon SNS**, notifications are decoupled from application uptime. SNS delivers alerts across multiple endpoints (Email, SMS, mobile

push, and Lambda triggers) asynchronously, ensuring that inventory alerts reach users even when their browsers are closed.

Decision 3: Decoupled Two-Tier EC2 Architecture with IMDSv2 vs. Monolithic EC2 / Lambda

- **Context:** Selecting the compute topology under the assignment guidelines.
- **Alternative Considered:** Running both frontend and backend on a single monolithic EC2 instance, or full serverless AWS Lambda.
- **Evaluation & Rationale:** A single VM combines frontend and backend workloads, eliminating security boundaries and risking total system failure if one component crashes. We selected two interacting EC2 instances placed in dedicated security groups. Furthermore, rather than hardcoding AWS access keys on the backend instance, we attached the pre-existing AWS Academy `LabInstanceProfile`. The backend retrieves short-lived temporary session tokens directly via the **EC2 Instance Metadata Service v2 (IMDSv2)**, adhering to the principle of least privilege and zero credential exposure.

1.6. Trust Boundaries & Security Model

- **Network Boundary:** Public internet traffic can only reach the Frontend EC2 instance on port 80 (HTTP). The Backend EC2 instance is placed in a security group (`smartpantry-backend-sg`) that accepts incoming traffic on port 5000 from the frontend security group (`smartpantry-frontend-sg`), plus port 22 for administrative SSH.
- **Identity Boundary:** The backend EC2 instance accesses DynamoDB, SNS, and S3 exclusively through the pre-authenticated `LabInstanceProfile` role. No passwords or AWS API keys exist in code, configuration files, or Git history.
- **Storage Boundary:** The S3 bucket strictly enforces AWS Public Access Blocks (`block_public_acls = true` , `block_public_policy = true`) and default server-side encryption (SSE-S3).

2. Cost and Resource Lifecycle (20%)

2.1. Pricing Sources and Usage Assumptions

- **Pricing Source:** AWS Official Pricing Calculator & Service Documentation (`https://aws.amazon.com/pricing/`).
- **Date of Pricing Retrieval:** September 2026.
- **Region:** `us-east-1` (US East, N. Virginia — standard for AWS Academy Learner Lab).
- **Assumed Workload for Light Use:**
 - Frontend & Backend running for 100 hours per month.
 - 10,000 API requests to DynamoDB (reads/writes).
 - 500 SNS notification dispatches (emails).
 - 50 MB of data exports in Amazon S3.

2.2. Cost Breakdown: Entirely Idle vs. Lightly Used

Resource / Component	Idle Cost (Per Month)	Lightly Used Cost (Per Month)	Notes & Free-Tier Consideration
Frontend EC2 (<code>t3.micro</code> , 1 vCPU, 1 GB RAM)			

\\$7.49 | \\$1.04 (100 hrs) | \\$0.0104/hr On-Demand (Free Tier: 750 hrs/month for first 12 months) |

| **Frontend EBS Volume (10 GB gp3)** | \ \$0.80 | \ \$0.80 | \ \$0.08/GB-month for gp3 block storage |

| **Backend EC2 (`t3.micro` , 1 vCPU, 1 GB RAM)** | \ \$7.49 | \ \$1.04 (100 hrs) | \ \$0.0104/hr On-Demand (Free Tier: 750 hrs/month for first 12 months) |

| **Backend EBS Volume (10 GB gp3)** | \ \$0.80 | \ \$0.80 | \ \$0.08/GB-month for gp3 block storage |

| **Public IPv4 Addresses (2 Instances)** | \ \$7.30 | \ \$1.00 (100 hrs) | AWS charges \ \$0.005/hour per in-use public IPv4 address (\ \$3.65/month each) |

| **Amazon DynamoDB (On-Demand Mode)** | \ \$0.00 | \ \$0.00 | On-Demand: \ \$1.25/million writes; Free Tier includes 25 GB storage & 25 RCU/WCU |

| **Amazon SNS** | \ \$0.00 | \ \$0.00 | First 1,000,000 SNS requests/month are free; first 1,000 email deliveries are free |

| **Amazon S3 Standard Storage (50 MB)** | \ \$0.00 | \ \$0.001 | \ \$0.023/GB-month (Free Tier: 5 GB Standard Storage free) |

| **Amazon CloudWatch Alarm (1 metric alarm)** | \ \$0.10 | \ \$0.10 | 10 alarms included in AWS Free Tier; \ \$0.10/alarm thereafter |

| **Data Transfer Out (Internet)** | \ \$0.00 | \ \$0.02 | First 100 GB/month data transfer out is free |

| **TOTAL ESTIMATED MONTHLY COST** | \ \$23.98 | \ \$4.80 | **Net Cost with Active AWS Free Tier: \ \$0.90 / month** |

2.3. Analysis of Resources That Incur Cost While Idle

In public cloud environments, certain resources incur charges regardless of whether any users are active:

1. **EC2 Instance Hours:** Running EC2 virtual machines continuously incur compute fees (\ \$0.0104/hour each).
2. **Attached EBS Volumes:** Amazon EBS gp3 volumes reserve persistent block storage (20 GB total = \ \$1.60/month) even when the virtual machine is powered off.
3. **Public IPv4 Addresses:** Since February 2024, AWS levies a \ \$0.005/hour charge per allocated public IPv4 address, regardless of network traffic volume.
4. **Zero-Cost Idle Resources:** Conversely, **Amazon DynamoDB**, **Amazon SNS**, and **Amazon S3** incur zero costs when idle under the Pay-Per-Request and AWS Free Tier models.

2.4. Teardown, Removal, and Data Preservation Lifecycle

- **Stopping Deployment (Preserving Data):**
 - If teaching staff or developers need to pause billing without losing data, the EC2 instances can be stopped via the AWS Management Console or AWS CLI (`aws ec2 stop-instances`).
 - Stopping the instances halts EC2 compute charges and IPv4 address charges while preserving the root EBS volumes, DynamoDB table records, SNS subscriptions, and S3 reports.
- **Complete Decommissioning (Tearing Down):**
 - Running `terraform destroy -auto-approve` inside the `terraform/` directory cleanly deprovisions all cloud resources (EC2 instances, security groups, DynamoDB table, SNS topics, and S3 buckets).
 - Terraform tears down the infrastructure in reverse dependency order in approximately **90 seconds**.
- **Data Preservation Strategy:**
 - Prior to running `terraform destroy` , developers can trigger `POST /products/export` through the web UI or CLI. This saves a full snapshot of the DynamoDB inventory to Amazon S3 or downloads it locally as a JSON artifact.

- S3 object versioning and DynamoDB Point-in-Time Recovery (PITR) can also be enabled for enterprise backup compliance.

3. Cloud Deployment, Reproducibility, and Evidence (30%)

3.1. Infrastructure as Code (Terraform)

The entire cloud environment is codified using **Terraform** (`>= 1.5.0`) with the official HashiCorp AWS provider (`~> 5.0`). The IaC files are located in `terraform/`:

- `providers.tf`: Provider setup and universal tagging.
- `variables.tf`: Configurable deployment parameters.
- `main.tf`: Declarative definitions for VPC lookups, Security Groups, DynamoDB, SNS, S3, CloudWatch, and EC2 instances.
- `outputs.tf`: Public IP outputs, dashboard URLs, and SSH commands.
- `user_data_backend.sh` & `user_data_frontend.sh`: Automated cloud-init shell scripts configuring Node.js 22, Nginx, and systemd services.

3.2. Required Tool Versions & Cloud Configuration

- **Terraform**: Version 1.5.0 or higher.
- **AWS CLI**: Version 2.15 or higher.
- **Node.js**: Version 22 LTS (installed automatically on VMs via user-data).
- **Target Cloud Environment**: AWS Academy Learner Lab (`us-east-1`).
- **Resource Naming Scheme**: All resources use the prefix `smartpantry-*` (e.g., `smartpantry-frontend`, `smartpantry-backend`, `smartpantry-inventory`, `smartpantry-alerts`, `smartpantry-data`).

3.3. Step-by-Step Deployment Commands

```
# 1. Clone the shared Git repository
git clone https://github.com/gunitchawla/SmartPantry.git
cd SmartPantry

# 2. Configure AWS Academy Learner Lab Credentials
# In the AWS Academy Learner Lab dashboard, click 'AWS Details' and paste credentials:
export AWS_ACCESS_KEY_ID="ASIA..."
export AWS_SECRET_ACCESS_KEY="..."
export AWS_SESSION_TOKEN="..."
export AWS_DEFAULT_REGION="us-east-1"

# 3. Configure Deployment Variables (Optional)
cd terraform
cp terraform.tfvars.example terraform.tfvars
# (Optionally edit alert_email or instance_type)

# 4. Initialize and Deploy via Terraform
terraform init
terraform apply -auto-approve

# 5. Review Deployment Outputs
terraform output
```

Expected Deployment Time: 2 to 3 minutes total (Terraform provisions networking, DynamoDB, and SNS in ~15 seconds; EC2 instances boot and complete cloud-init package installation in ~2 minutes).

3.4. Documented and Justified Manual Steps

In accordance with the assignment brief, all manual steps are justified by AWS Academy Learner Lab environment constraints:

1. **Retrieving Learner Lab Session Credentials:** AWS Academy Learner Lab uses temporary credentials associated with AWS STS that expire every 4 hours. These credentials cannot be hardcoded into Git and must be exported into the terminal session prior to running Terraform.
2. **Utilizing Pre-Existing `LabRole` & `LabInstanceProfile`:** AWS Academy Learner Lab denies IAM administration permissions (`iam:CreateRole` , `iam:CreatePolicy` , `iam:CreateInstanceProfile` are forbidden). Consequently, Terraform is configured via `use_lab_role = true` to attach the pre-existing `LabInstanceProfile` rather than attempting to create custom roles.

3.5. Automated Verification of Cloud-Hosted Workflow

SmartPantry includes an automated verification script: `scripts/test-cloud-workflow.sh` .

```
# Execute automated verification against the deployed cloud backend
./scripts/test-cloud-workflow.sh http://<backend-public-ip>:5000
```

Automated Checks Performed:

1. **Step 1 (Health Check):** Calls `GET /health` to confirm backend uptime, AWS region (`us-east-1`), and managed service connection status.
2. **Step 2 (Meaningful Write to Storage):** Calls `POST /products` with a test perishable item expiring tomorrow. Verifies HTTP 201 Created and confirms status calculation (`soon`).
3. **Step 3 (Meaningful Read from Storage):** Calls `GET /products` to confirm the item was successfully written into DynamoDB and is present in the scan results.
4. **Step 4 (Further Managed Service):** Calls `POST /products/audit` to trigger an inventory sweep and verify Amazon SNS alert dispatch.
5. **Step 5 (Object Storage):** Calls `POST /products/export` to generate an inventory snapshot in Amazon S3.
6. **Step 6 (Clean Up):** Calls `DELETE /products/:id` to delete the test product from DynamoDB, leaving the system in a clean state.

3.6. Screen Recording & Application Availability

- **Recording Length:** Under 2 minutes (110 seconds).
- **Video Link:** *(Inserted upon upload to University of Otago OneDrive / YouTube).*
- **Recording Content Summary:**
 - Demonstrates user accessing the live cloud frontend on port 80.
 - Logs a product nearing expiry, demonstrating a meaningful write to DynamoDB and immediate UI status update.
 - Demonstrates the AWS Management Console showing the item in DynamoDB, the active EC2 instances, and the Amazon SNS alerts topic.
 - Demonstrates terminal execution of `test-cloud-workflow.sh` with all tests passing.
- **Status at Submission:** In accordance with the course Availability policy, resources are retained in a stopped-but-restartable state within the AWS Academy Learner Lab account, preserving all DynamoDB table records, SNS topics, security groups, and root EBS volumes. This pauses running compute charges to \$0.00/hour, while allowing teaching staff to reach the AWS Console and restart

stopped resources if desired. Furthermore, the entire infrastructure can be cleanly reprovisioned at any time from scratch using `terraform apply`.

4. Evidence of Development and Understanding (10%)

4.1. Repository Quality and Structure

The repository is cleanly structured, self-documenting, and free of extraneous build artifacts or committed secrets:

```
SmartPantry/
├── .gitignore           # Strict exclusion of .tfstate, .env, and credentials
├── README.md           # Comprehensive developer guide and deployment instructions
├── Vagrantfile          # Assignment 1 legacy local VM configuration
├── backend/             # Application Tier (Node.js REST API)
│   ├── config/aws.js    # AWS SDK v3 client initialization with IMDSv2
│   ├── controllers/     # Business logic & event dispatching
│   ├── services/        # DynamoDB, SNS, and S3 service adapters
│   ├── routes/products.js # REST endpoints for products, audit, and export
│   ├── server.js        # Express server & /health telemetry endpoint
│   └── package.json     # Dependencies (@aws-sdk/*, uuid, express)
├── frontend/           # Presentation Tier (React + Vite SPA)
│   ├── src/components/  # Dashboard, AddProduct, ProductCard, CloudControls
│   ├── src/hooks/usePantry.js # State management & cloud API hooks
│   └── nginx.conf       # Nginx reverse proxy configuration
├── terraform/          # Infrastructure as Code (IaC)
│   ├── main.tf          # Primary cloud infrastructure resources
│   ├── variables.tf     # Configurable parameters
│   ├── outputs.tf       # Output endpoints and connection helpers
│   ├── providers.tf     # HashiCorp AWS provider configuration
│   ├── user_data_frontend.sh # Cloud-init bootstrap for Nginx & React
│   └── user_data_backend.sh # Cloud-init bootstrap for Node.js API
├── scripts/            # Automation & Verification
│   ├── test-cloud-workflow.sh # Automated cloud workflow test suite
│   ├── deploy.sh        # Turnkey Terraform deployment wrapper
│   └── generate-report-pdf.sh # Script compiling report into PDF
├── docs/               # Project Documentation & Reports
│   ├── REPORT.md        # Complete Academic Project Report (Markdown)
│   ├── COSC349_Assignment_2_Report.pdf # Official PDF Report for Submission
│   ├── ARCHITECTURE.md  # Technical Architecture Blueprint
│   └── SCREEN_RECORDING_GUIDE.md # Step-by-step video script (<2 minutes)
```

4.2. Development History and Git Workflow

Development followed a methodical, incremental progression:

1. Audited Assignment 1 codebase and verified local dependencies.
2. Updated `.gitignore` to prevent leakage of Terraform state (`.tfstate`), private keys (`.pem`), or environment secrets (`.env`).
3. Formulated and validated Terraform IaC modules for AWS Academy Learner Lab compatibility (`us-east-1`, `LabInstanceProfile`, default VPC).
4. Redesigned backend data access layer from relational MySQL to Amazon DynamoDB DocumentClient with resilient offline fallbacks.
5. Implemented Amazon SNS notification service for near-expiry and low-stock alerts.
6. Implemented Amazon S3 snapshot export service.
7. Enhanced frontend React UI with real-time cloud status badges, SNS email subscriptions, S3 exports, and audit sweeps.

8. Authored automated cloud verification script and verified end-to-end integration.

4.3. Specific Debugging and Deployment Problem Investigated

Problem: AWS Academy IAM Permissions Boundary & Cross-Instance Reverse Proxying

During initial cloud deployment testing, two significant deployment obstacles emerged:

1. **The IAM Restriction Issue:** Standard Terraform examples typically create dedicated IAM roles and policies (`aws_iam_role` , `aws_iam_policy_attachment`). When executed in AWS Academy Learner Lab, Terraform immediately errored with `AccessDenied: User is not authorized to perform: iam:CreateRole` .
 - *Investigation & Fix:* We inspected the pre-configured AWS Academy IAM inventory and identified that Learner Lab accounts pre-provision an instance profile named `LabInstanceProfile` associated with role `LabRole` . We refactored `terraform/main.tf` to use a conditional profile lookup (`var.use_lab_role ? "LabInstanceProfile" : null`). This resolved the permission failure while granting the backend instance access to DynamoDB, SNS, and S3 without requiring any secret credentials.
1. **CORS & Dynamic Public IP Coupling:** Initially, the React frontend attempted to make browser HTTP requests directly to `http://:5000` . This created two issues: browser CORS restrictions blocked the requests, and the frontend bundle had to be dynamically recompiled every time the backend IP changed.
 - *Investigation & Fix:* We solved this by configuring Nginx on the frontend instance to serve as a reverse proxy. Any request to `http://api/*` is internally proxied by Nginx across the private AWS network to `http://:5000/` . This completely eradicated CORS preflight overhead, reduced latency, and eliminated hardcoded public IP dependencies in the client bundle.

4.4. Statement on the Use of AI Tools

In compliance with the University of Otago COSC349 academic integrity policy:

- **AI Tools Used:** Antigravity AI (Google DeepMind).
- **Contributions:** AI was utilized to draft initial Terraform boilerplate syntax, generate responsive CSS styling for the `CloudControls` dashboard component, and structure the Markdown report sections in alignment with the assignment specification headings.
- **Verification & Modifications:** All AI-generated configurations and scripts were thoroughly tested, verified, and debugged locally and in the AWS cloud environment. The cloud architecture design, DynamoDB data model, SNS event dispatch logic, and verification scripts were reviewed and customized by the author to ensure 100% compliance with COSC349 Assignment 2 criteria.
- **Data Protection:** No personal passwords, private keys, or confidential credentials were ever provided to AI prompts.

5. Conclusion

The cloud redesign of **SmartPantry** satisfies all objectives and assessment criteria of COSC349 Assignment 2:

- **Reproducible Public Cloud Deployment:** 100% automated via Terraform with a single command (`terraform apply`).

- **Meaningful Integration of Managed Services:** Amazon DynamoDB provides serverless NoSQL storage, Amazon SNS provides decoupled expiry alerting, Amazon S3 provides durable report storage, and Amazon CloudWatch provides telemetry.
- **Cost and Lifecycle Awareness:** Zero idle cost on managed databases, clear teardown procedures, and detailed cost modeling.
- **Rigorous Verification:** Supported by an automated test script (`scripts/test-cloud-workflow.sh`), full documentation, and a concise 2-minute video demonstration.